

Unit 4, Lesson 1: Algebraic Descriptions for Translations and Rotations



MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.



- ☐ I can relate a translation on the coordinate plane to an algebraic notation of the transformation.
- ☐ I can relate a rotation about the origin on the coordinate plane to an algebraic notation of the transformation.



A career as a pharmacist is just one of many jobs in the health care field. Pharmacists need an understanding of both science and mathematics. Pharmacists also need to work with patients on their prescriptions to ensure the medication is working as it should.



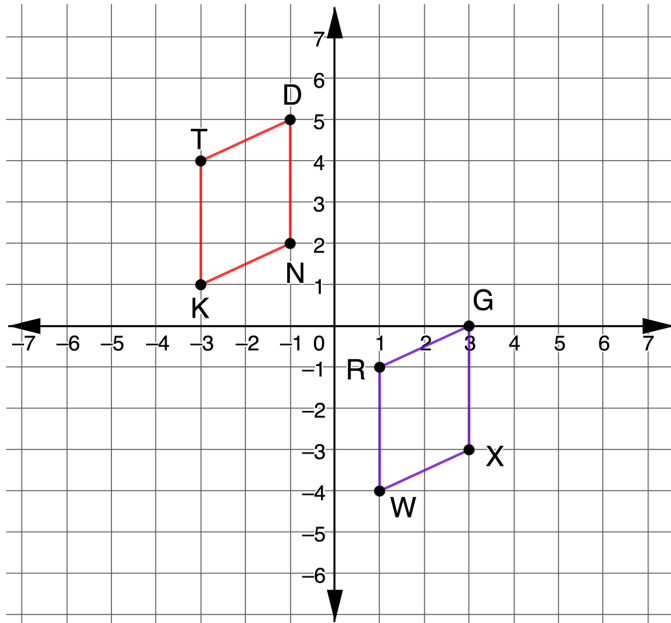
1. What is one thing you learned about pharmacists that you did not know?
2. How does paying attention to detail help you to learn mathematics?





4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations

1. Parallelograms $DNKT$ and $GXWR$ are shown on the coordinate grid.



- a. Describe the transformation if parallelogram $DNKT$ is the preimage.
- b. Describe the transformation if parallelogram $GXWR$ is the preimage.

- c. Complete the table by writing the x -value of each vertex in the columns provided.

Then, determine the number of units that should be added or subtracted to the x -value for the vertices of parallelogram $DNKT$ to get the x -value for the vertices of parallelogram $GXWR$.

Vertex	x	Vertex	x	Description
D		G		☐ Add ☐ Subtract _____
N		X		☐ Add ☐ Subtract _____
K		W		☐ Add ☐ Subtract _____
T		R		☐ Add ☐ Subtract _____

- d. Complete the table by writing the y -value of each vertex in the columns provided.

Then, determine the number of units that should be added or subtracted to the y -value for the vertices of parallelogram $DNKT$ to get the y -value for the vertices of parallelogram GXR .

Vertex	y	Vertex	y	Description
D		G		☐ Add ☐ Subtract
N		X		☐ Add ☐ Subtract
K		W		☐ Add ☐ Subtract
T		R		☐ Add ☐ Subtract

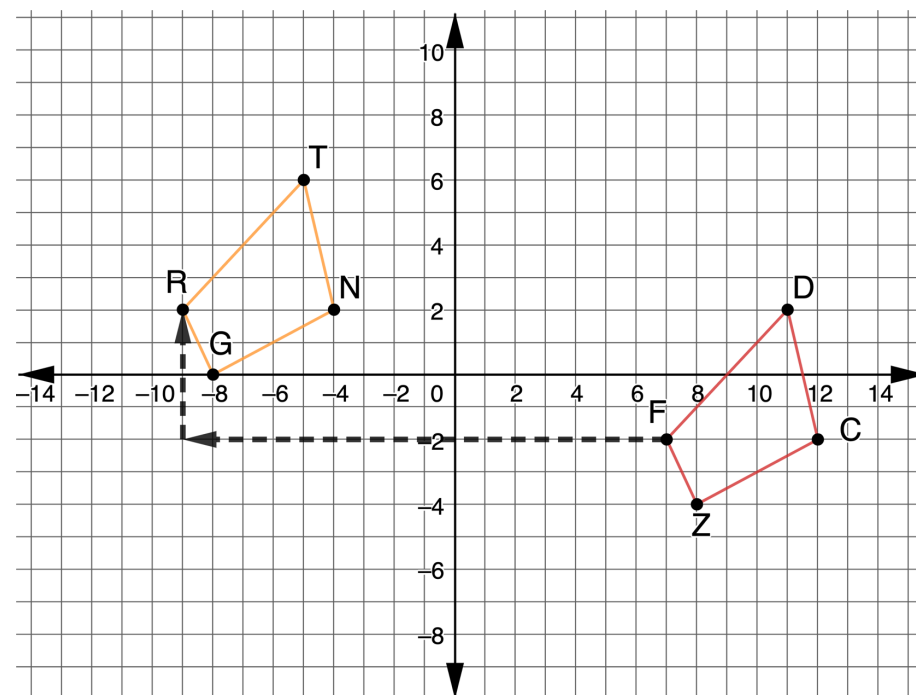
Transformations can be described using words, as well as using a notation.

- e. Use the tables to complete the algebraic description of the transformation if parallelogram $DNKT$ is the preimage.

$$(x, y) \rightarrow (x \boxed{}, y \boxed{})$$

- f. Discuss with your partner how the algebraic description of the transformation would change if parallelogram GXR is the preimage.

2. Quadrilaterals $GRTN$ and $ZFDC$ are shown on the coordinate grid. A mapping of vertex F is shown.

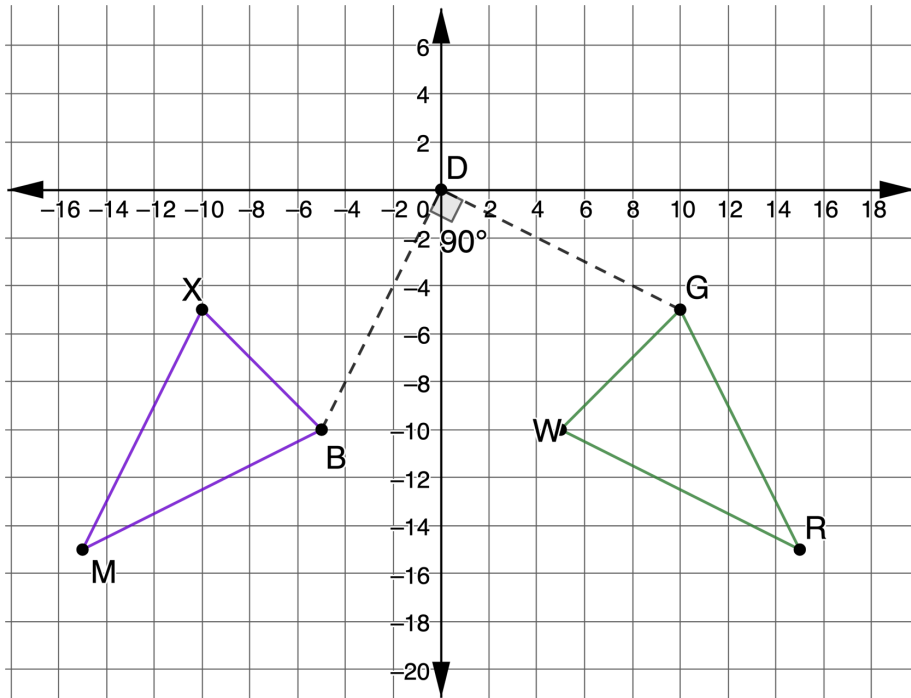


- a. Map the transformation for the other vertices of quadrilateral $ZFDC$.
- b. Discuss with your partner how the algebraic description of the transformation, $(x, y) \rightarrow (x - 16, y + 4)$, is revealed in the mapping.



4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations

1. Triangles BXM and GWR are shown on the coordinate grid, where $\triangle GWR$ is the preimage. A mapping of vertex G is shown.



- a. Map the transformation for the other vertices of triangle GWR .
- b. Describe the transformation if triangle GWR is the preimage.

- c. Discuss with your partner how to use the table of values to describe the relationship of the corresponding vertices using an algebraic description.

Triangle GWR

Vertex	x	y
G	10	-5
W	5	-10
R	15	-15

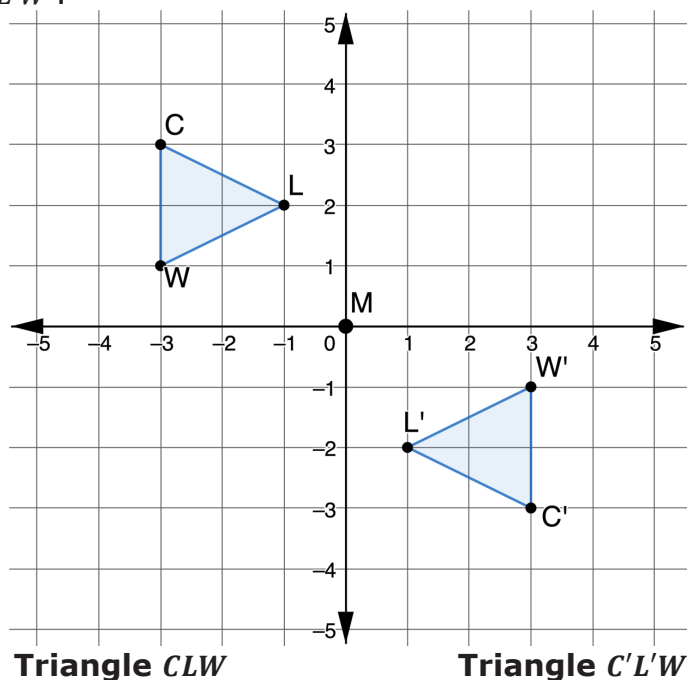
Triangle BXM

Vertex	x	y
B	-5	-10
X	-10	-5
M	-15	-15

- d. Write a conjecture that you and your partner think will work for the algebraic description of the rotation.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

2. Triangles CLW and $C'L'W'$ are shown on the coordinate grid. Triangle CLW was rotated 180° about M to create $\Delta C'L'W'$.



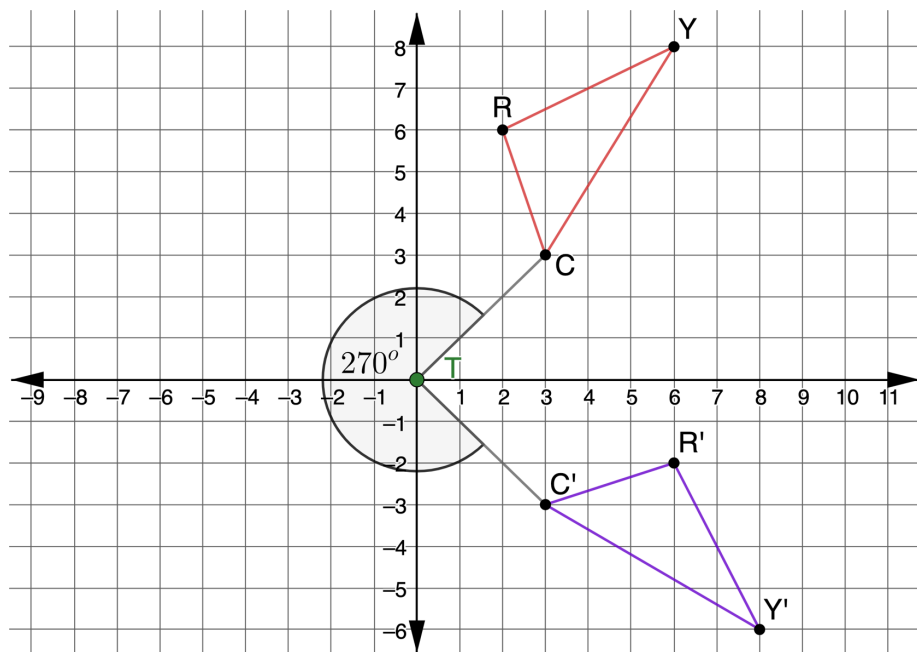
Vertex	x	y
C	-3	3
L	-1	2
W	-3	1

Vertex	x	y
C'	3	-3
L'	1	-2
W'	3	-1

- a. It was stated that ΔCLW was rotated 180° about M to create $\Delta C'L'W'$. Was that rotation clockwise or counterclockwise?

- b. Discuss with your partner how your previous conjecture works with rotating ΔCLW 180° about M . If necessary, write a new conjecture that works for rotations of 180° about the origin. Include the direction of the rotation determined.
- c. Assuming $\Delta C'L'W'$ was the preimage and is rotated 180° about the origin, test whether your previous or new conjecture works to describe this rotation. Include the direction of rotation.

3. Triangles YCR and $Y'C'R'$ are shown on the coordinate grid. $\triangle YCR$ was rotated 270° counterclockwise about T to create $\triangle Y'C'R'$.



Triangle YCR

Vertex	x	y
Y	6	8
C	3	3
R	2	6

Triangle $Y'C'R'$

Vertex	x	y
Y'	8	-6
C'	3	-3
R'	6	-2

- a. Discuss with your partner how the ordered pairs for $\triangle YCR$ and $\triangle Y'C'R'$ could be used to write a conjecture that works for rotations of 270° counterclockwise about the origin.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

- b. Test your conjecture for rotations of 270° counterclockwise by looking at $\triangle BXM$ and $\triangle GWR$ from question 1. Assume $\triangle GWR$ is the preimage and was rotated 270° counterclockwise about the origin. If necessary, adjust your conjecture.
- c. Discuss with your partner why the same conjecture should work for a rotation of 270° counterclockwise and 90° clockwise about the origin.



4.1.4 Bring It Together: Algebraic Descriptions for Translations and Rotations

Translations	
Right	Left
Add to the x -coordinate $(x, y) \rightarrow (x + \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	Subtract from the x -coordinate $(x, y) \rightarrow (x - \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
Up	Down
Add to the y -coordinate $(x, y) \rightarrow (\underline{\hspace{1cm}}, y + \underline{\hspace{1cm}})$	Subtract from the y -coordinate $(x, y) \rightarrow (\underline{\hspace{1cm}}, y - \underline{\hspace{1cm}})$

Rotations clockwise about the origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
Rotations counterclockwise about the origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

1. What do you notice about the algebraic descriptions for rotations?



4.1.5 Wrap-Up: Cool Down

1. A student asks you for help with algebraic descriptions for rotations because they are stuck on how to do it. Describe the process using what you learned in this lesson, highlighting any potential misconceptions they may have.